INCH-POUND

MIL-S-20708/48E
13 March 1989

SUPERSEDING
MIL-S-20708/48D
30 July 1987

MILITARY SPECIFICATION SHEET

SYNCHRO TORQUE DIFFERENTIAL TRANSMITTER, TYPE 23TDX4C

This specification is approved for use by all Departments and Agencies of the Department of Defense. The requirements for acquiring the Synchros described herein shall consist of this specification and the latest issue of MIL-S-20708

TABLE I Requirements

Requirement	Value	Unit	Tolerance
Frequency	400	Hz	±1%
Primary Voltage	78 Ⓔ	volts	±1%
Primary Current	866 0	milliamps	maximum
Primary Power	7 0	watts	maximum
Impedance			
Zso	90-112	ohms	min -max
Zrs	11 0-15 0	ohms	min -max
Impedance Angle			
Zso	84 0-88 0	degrees	min -max
Zrs	47 0-53 0	degrees	min -max
Transformation Ratio	1 154	------	±2%
Phase Shift (Lead)	2 0	degrees	±1 0
Electrical Error	8 0	minutes	maximum
Torque Gradient	0 16	ounce-inches per degree	minimum
Friction Torque	0 20	ounce-inches	maximum
Radial Play	0 0010	inches	maximum
End Play	0 0025	inches	±0 0020
Temperature Rise	25 0	degrees (C)	maximum
Variation of Voltage (+10%) and Freq (-5%)	7 84	watts	maximum

Ⓔ denotes changes

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TABLE II Military part number variant characteristics

Military Part No 1/	A ± 010	B ± 010	G Maximum	Shaft Style	Terminal End	U + 0000 - 0002
M20708/48-01C	676	719	3 410	Splined	Terminal Board	2405

(E)

1/Military part number M20708/48-01C shall be the NATO standard

1/The suffix letter following the numerical dash number corresponds to the latest modification letter in the type designation

TABLE III Military part number cross-references

SUPERSEDED MILITARY PART NUMBERS		NEW MILITARY PART NUMBERS
MIL-S-20708/48C	MIL-S-20708/48B	
M20708/48C-001	M20708/48-001	M20708/48-01C

NOTE All line items shown in Table III refer to equivalent and interchangeable synchros of the same type designation modification Part number changes do not affect form, fit, or function of the synchros listed therein

Custodians
Army-AR
Navy-AS
Air Force-85

Review Activities
Army-MI,AV
Navy-OS
Air Force-11,17,99
DLA-ES

User Activities
Navy-MC,CG

Preparing Activity
Navy-AS

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